

LAB MANUAL

Experiment: Project Management Tool using Trello

Aim:

To understand the basics of project management and to plan, track, and manage a software project using a project management tool (Trello).

Objectives:

1. To understand project management concepts
2. To use a project management tool
3. To create and manage project tasks
4. To track project progress using workflow boards

Theory:

Project management involves planning, organizing, and managing resources to successfully complete project goals. Project management tools help teams collaborate, assign tasks, monitor progress, and meet deadlines efficiently.

Trello is a Kanban-based project management tool that uses boards, lists, and cards to represent projects and tasks visually.

Software Requirements:

- Web Browser (Chrome/Edge/Firefox)
- Internet Connection
- Trello Account

Procedure:

- Step 1: Open a web browser and go to trello.com
- Step 2: Login or sign up using your email ID
- Step 3: Click on Create → Create Board
- Step 4: Name the board as "Student Management System"
- Step 5: Create lists: To Do, In Progress, Testing, Done
- Step 6: Add task cards under To Do list
- Step 7: Assign due dates and members to cards
- Step 8: Move cards across lists as task progress changes

Part A: Agile Setup

1. Create a new board named “Agile Project – Student Management System”
2. Create lists:

- Product Backlog
 - Sprint Backlog
 - In Progress
 - Testing
 - Done
3. Add Product Backlog cards:
- User Login Module
 - Student Registration
 - Course Management
 - Attendance Tracking
 - Result Generation
4. Add user story description:
As a user, I want ____ so that ____
5. Move selected cards to **Sprint Backlog**
6. Add checklist to Sprint cards:
- Design
 - Coding
 - Unit Testing
 - Integration
7. Assign sprint due date, members, and sprint label

Part B: Automation using Butler

8. Click **Automation** → **Rules**
9. Create rule:
- When all checklist items are completed → Move card to **Done**
10. Create rule:
- When due date is marked complete → Move card to **Done**
11. Create rule:

- When label “Bug” is added → Move card to **Testing**

Part C: Monitoring

12. Enable **Calendar Power-Up**
13. Review completed sprint tasks
14. Conduct Sprint Retrospective

Expected Output

An Agile Scrum board with automated task movement and sprint visualization.

Expected Output:

A Trello board displaying tasks categorized under different workflow stages with progress tracking.

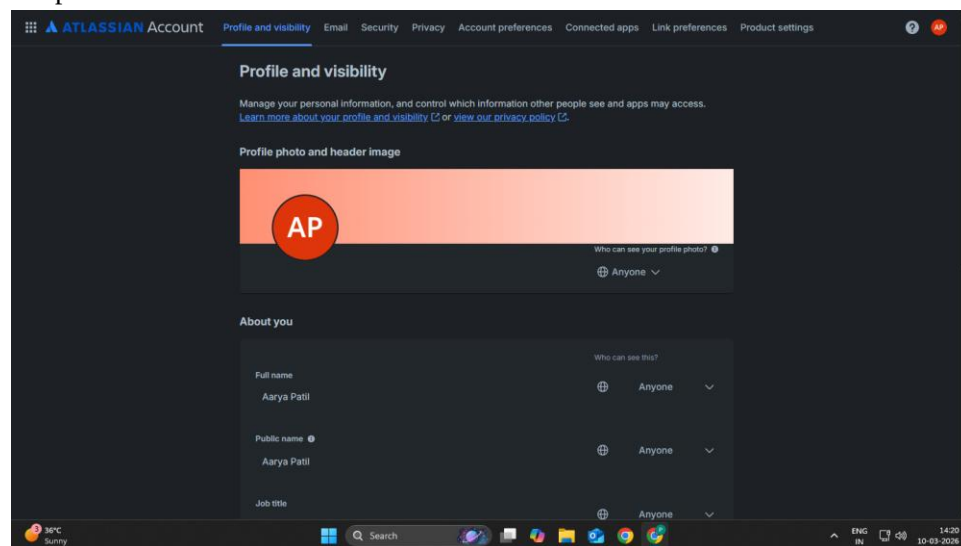
Result:

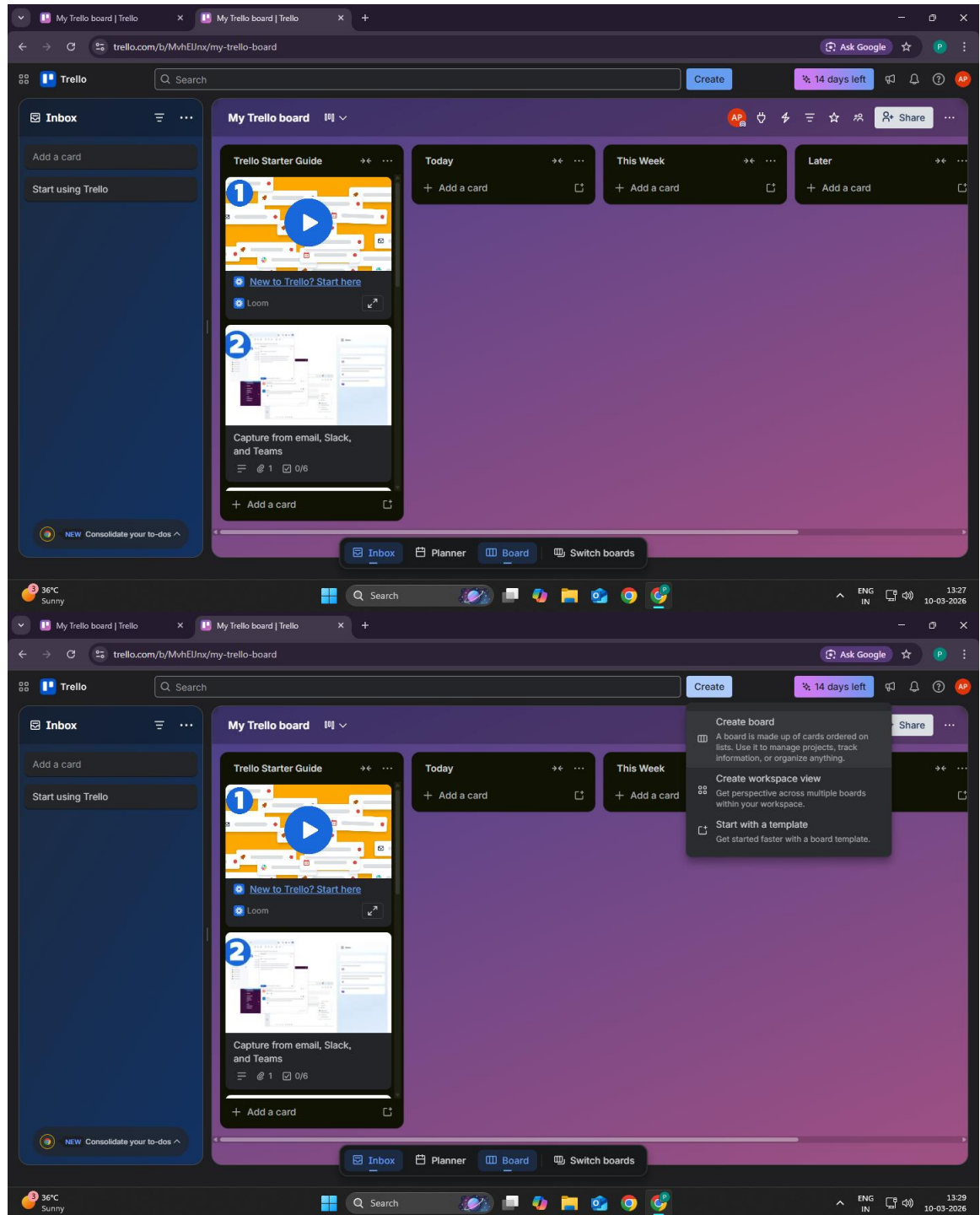
Thus, the project management tool Trello was successfully used to plan and manage a software project.

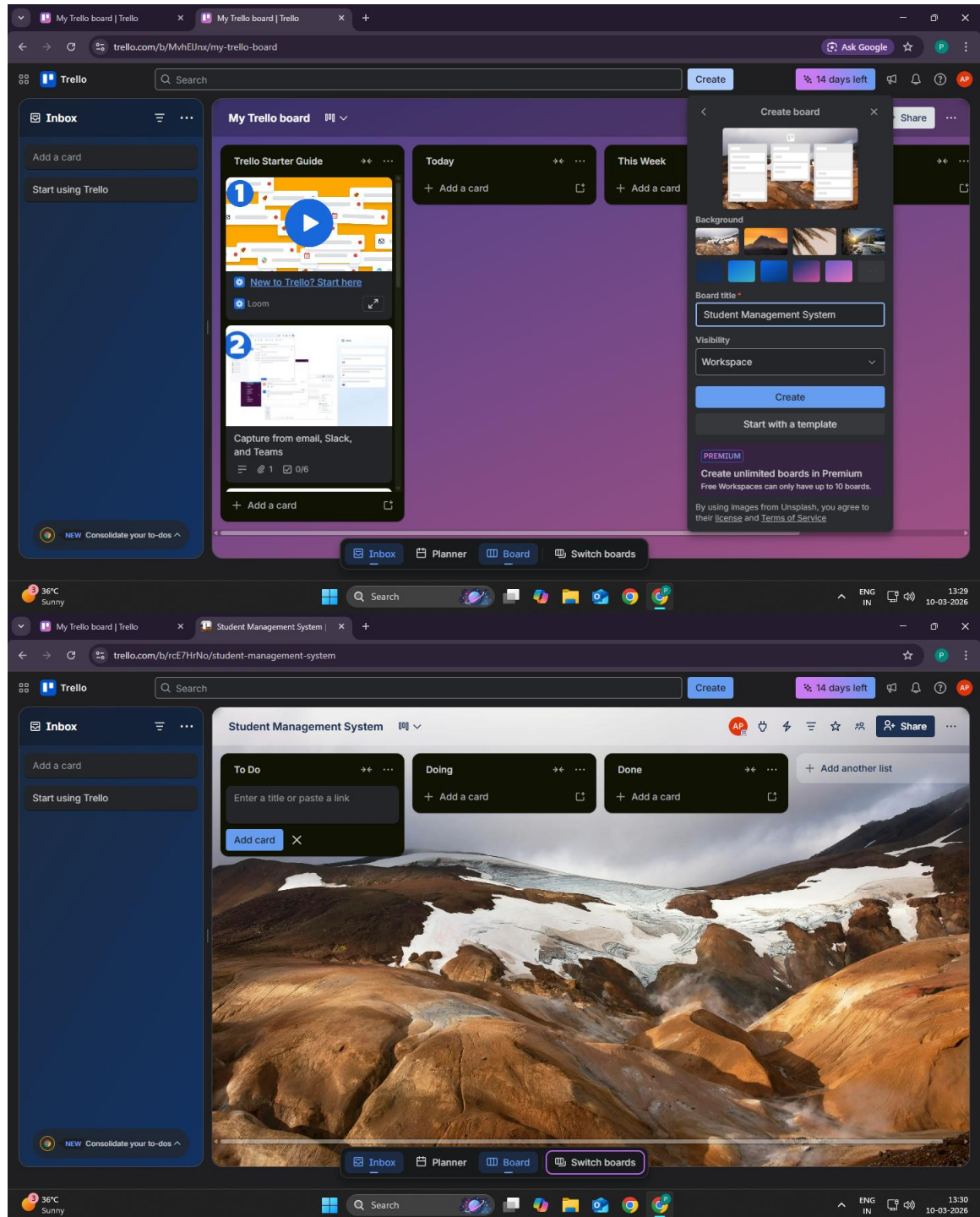
Conclusion:

1. What is project management?
2. What is Trello?
3. What is Kanban?
4. Advantages of project management tools
5. Difference between task and milestone

Output:







The screenshot displays a Trello board titled "Student Management System" in a web browser. The board is organized into four columns: "To Do", "In Progress", "Testing", and "Done". The "To Do" column contains a list of tasks: "User Login Module", "Student Registration", "Course Management", "Attendance Tracking", and "Result Generation". The "User Login Module" card is selected, opening a detailed view. This view includes a "Description" field with the text: "As a user, I want to log in to the system using my username and password so that I can securely access my student dashboard and use the system features." Below the description are buttons for "Save", "Discard changes", and "Formatting help". To the right of the description is a "Comments and activity" section showing a comment from "Aarya Patil" stating "Aarya Patil added this card to To Do" 1 minute ago. The browser's address bar shows the URL "trello.com/b/cE7HrNo/student-management-system". The Windows taskbar at the bottom indicates the system time as 13:49 on 10-03-2026.

The screenshot displays a Trello board titled "Student Management" with a "To Do" list. The first card, "User Login Module", is selected, showing its details. The card's description is "As a user, I want to log in to the system and use the system features, that I can securely access my student dashboard and use the system features." The card has a due date of "Mar 17, 1:53 PM" and is assigned to "Aarya Patil". A date picker is open, showing the month of March 2026. The card also has a "Comments and activity" section with a comment from "Aarya Patil" stating "Aarya Patil added this card to To Do 5 minutes ago". The board's left sidebar lists other cards: "User Login Module", "Student Registration", "Course Management", "Attendance Tracking", and "Result Generation". The bottom of the screen shows a Windows taskbar with the date "10-03-2026" and time "13:53".

Student Management To Do

User Login Module

+ Add Labels

Description

As a user, I want to log in to the system and use the system features, that I can securely access my student dashboard and use the system features.

Dates

March 2026

Start date

Due date

3/17/2026 1:53 PM

Recurring NEW

Never

Set due date reminder

1 Day before

Reminders will be sent to all members and watchers of this card.

Save

Comments and activity

Show details

Write a comment...

AP Aarya Patil added this card to To Do 5 minutes ago

Members

Due date

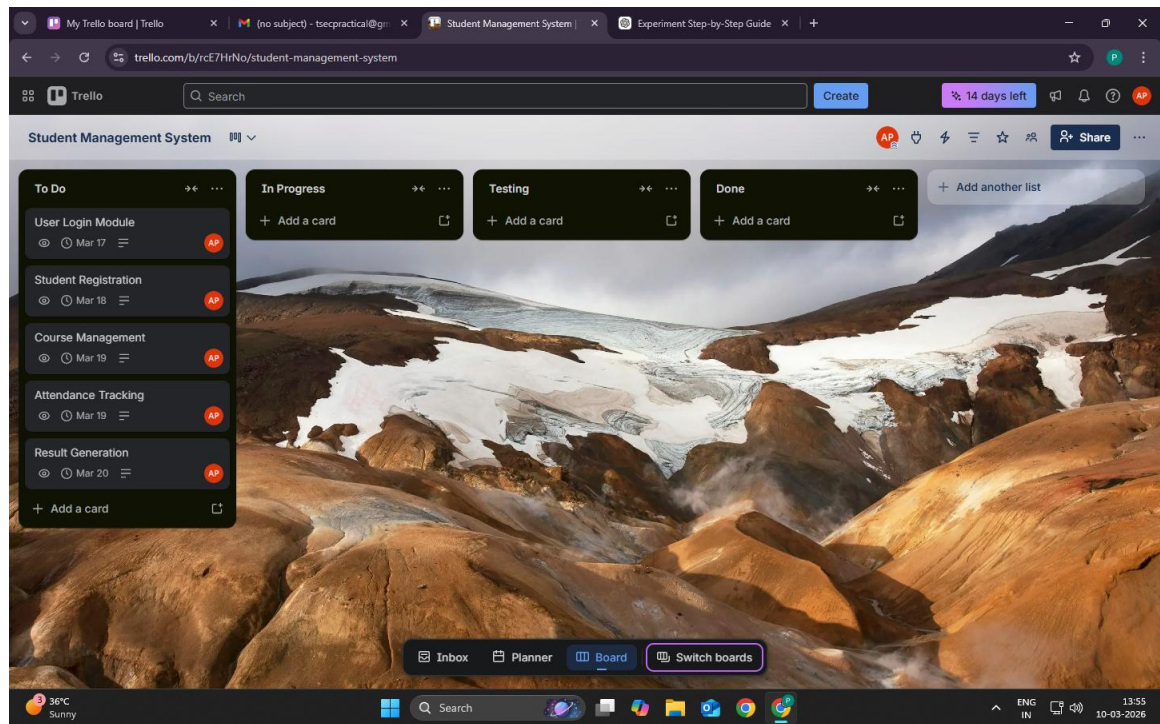
Mar 17, 1:53 PM

Search members

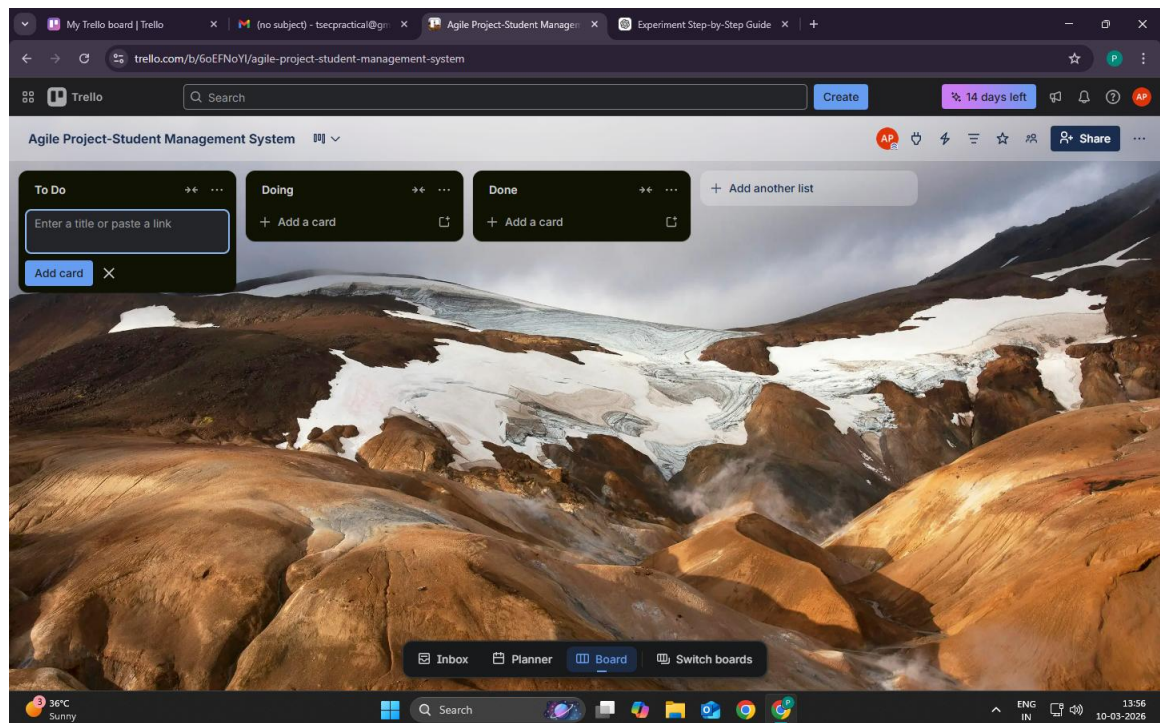
Card members

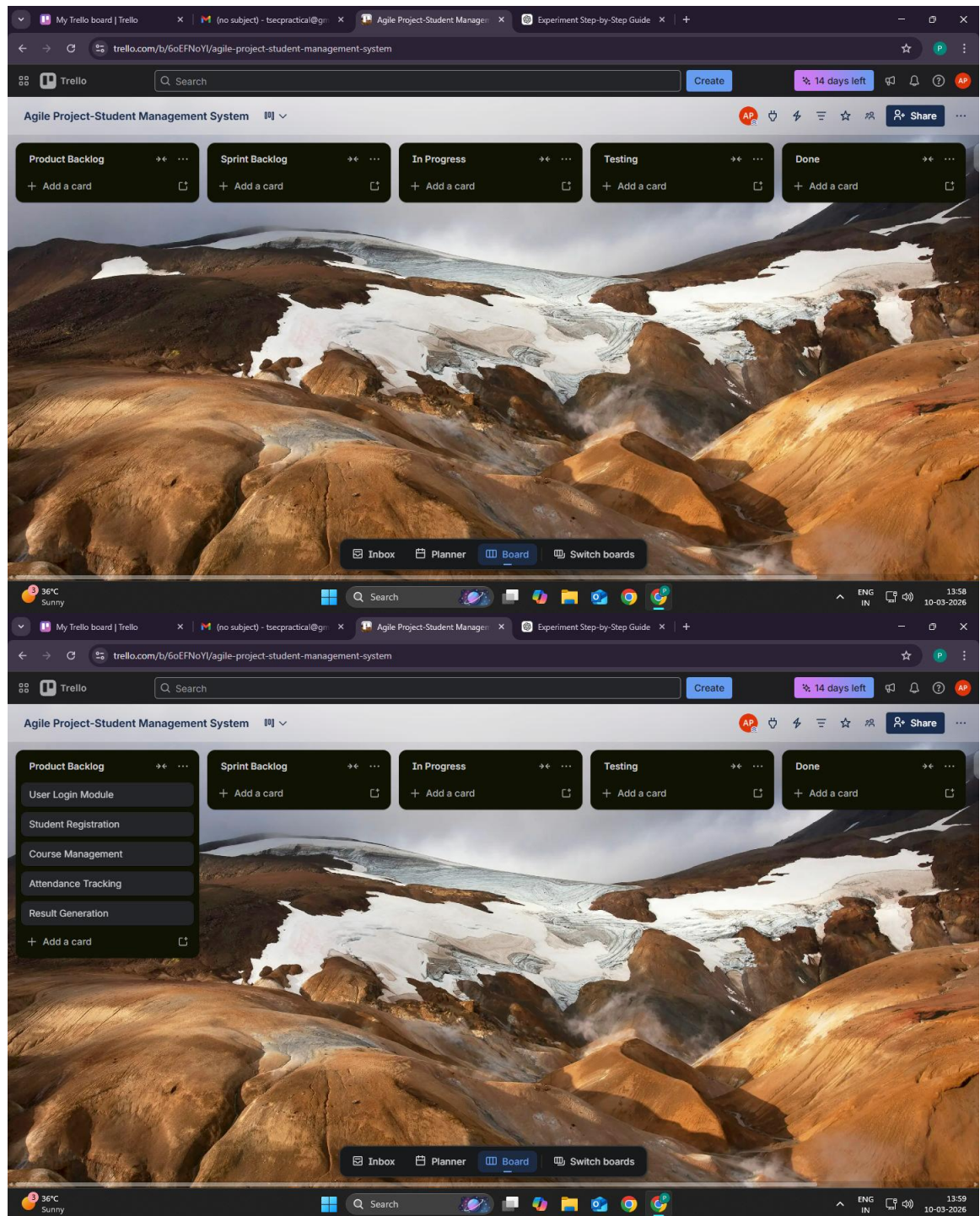
AP Aarya Patil

Power-ups Automations Comments



Part A:





The image shows a Trello board titled 'Agile Project-Studer' with a 'Sprint Backlog' filter. A card titled 'User Login Module' is selected, displaying a modal for adding a checklist. The modal includes a 'Title' field with 'Development Steps' and an 'Add' button. The background card shows a description field and a checklist with items: Design, Coding, Unit Testing, and Integration. A comment on the right states 'Aarya Patil added this card to Product Backlog 1 minute ago'. The interface includes a sidebar with navigation links like 'Product Backlog', 'Course Management', and 'Attendance Tracking'. The bottom of the screen shows a Windows taskbar with the date 10-03-2026 and time 14:01.

My Trello board | Trello

(no subject) - tscpractical@gm... User Login Module on Agile P... Experiment Step-by-Step Guide

trelllo.com/c/sQrqoYsA/1-user-login-module

Trello

Agile Project-Studer Sprint Backlog

Product Backlog

Course Management

Attendance Tracking

Result Generation

+ Add a card

User Login Module

+ Add Labels Dates Checklist Members

Description

Add a more detailed description...

Add checklist

Title

Development Steps

Add

Comments and activity

Show details

Write a comment...

Aarya Patil added this card to Product Backlog 1 minute ago

Power-ups Automations Comments

36°C Sunny

Search

ENG IN 14:00 10-03-2026

My Trello board | Trello

(no subject) - tscpractical@gm... User Login Module on Agile P... Experiment Step-by-Step Guide

trelllo.com/c/sQrqoYsA/1-user-login-module

Trello

Agile Project-Studer Sprint Backlog

Product Backlog

Course Management

Attendance Tracking

Result Generation

+ Add a card

User Login Module

+ Add Labels Dates Checklist Members

Description

Add a more detailed description...

Development Steps Delete

0%

Design

Coding

Unit Testing

Integration

Add an Item

Comments and activity

Show details

Write a comment...

Aarya Patil added this card to Product Backlog 1 minute ago

Power-ups Automations Comments

36°C Sunny

Search

ENG IN 14:01 10-03-2026

The image shows a Trello board titled "Agile Project-Student Management System". The board is organized into columns: "Product Backlog", "Sprint Backlog", "In Progress", "Testing", and "Done". The "Sprint Backlog" column contains a card titled "User Login Module". This card has three labels: "Sprint 1" (yellow), "Bug" (red), and "Feature" (purple). A comment by Aarya Patil is visible on the card, stating "Aarya Patil added this card to Product Backlog 1 minute ago". The board also shows a "Labels" panel on the left with a search bar and a list of labels. The bottom of the screen shows a Windows taskbar with various application icons and a system tray displaying the date and time.

Agile Project-Student Management System

Product Backlog

- Course Management
- Attendance Tracking
- Result Generation
- + Add a card

Sprint Backlog

- User Login Module
- Student Registration
- + Add a card

In Progress

- + Add a card

Testing

- + Add a card

Done

- + Add a card

Labels

- Sprint 1
- Bug
- Feature

Comments and activity

Aarya Patil added this card to Product Backlog 1 minute ago

Part B:

The image displays two screenshots of the Trello 'Rules' interface, illustrating the process of creating an automation rule.

Top Screenshot: Rules Overview

- Automation Section:** Includes links for Automations, Rules, Scheduled, Due date, Custom buttons, Card buttons, Board buttons, App automations, View apps, More, Suggestions, Activity, Send feedback, and Help.
- Rules Section:** Features a 'Create automation' button, 'Libraries', and 'Admin' options.
- Rules Description:** States 'Rules are simple: when one thing happens, another thing happens automatically'.
- Examples:**
 - When a card is created in list "To Do" by me, add the "Steps" checklist.
 - When a card is moved to list "Done" by anyone, mark the due date as complete and remove all members from the card.
 - When I am added to a card, set the due date in 5 working days and post a comment saying "I got this!"
- Create automation button:** A blue button to start creating a new rule.
- Video Overview:** A video thumbnail titled 'How to create board rule automation in Trello' with a play button and the text 'Watch a 65-second overview of rules'.

Bottom Screenshot: Create a Rule Dialog

- Create a Rule Header:** Includes a 'Watch video overview' link, 'Save', and 'Cancel' buttons.
- Condition:** A text box containing 'is completed in a card by me'.
- Select Action:** A section with various action icons: Move, Add/Remove, Dates, Checklists, Members, Content, Fields, Sort, Cascade, Jira, Bitbucket, and Slack.
- Action List:**
 - move the card to the top of list (with a dropdown menu showing 'List name' and a plus icon).
 - move the card to the top of the list (with a plus icon).
 - archive the card (with a plus icon).

